

Uncertainty-Aware Scoring Pre-Test

Executive Summary

The crude ensemble is biased structural perturbation, not a posterior, and a positive result only licenses building the real conditional-seed ensemble; it does not itself demonstrate that uncertainty-aware selection works.

The pre-test is weak/ambiguous: heteroscedasticity signals appear, but the sigma term does not convert into a strict selection gain.

The decision uses mild strictly and requires the same direction on mid. wide is reported only as a contaminated sanity bound; mild_no_full is a labeled non-decision diagnostic.

Framing Caveats

- The crude ensemble is biased structural perturbation, not a posterior, and a positive result only licenses building the real conditional-seed ensemble; it does not itself demonstrate that uncertainty-aware selection works.
- The only non-destructive same-architecture scorings are full and full_rescore; in this run they are deterministic, so ablation spread is structural damage rather than bf16 jitter.
- The enriched Books set is paper-pipeline-selected, not an independent gold standard, so enriched retention is a calibration diagnostic rather than a standalone proof of selection quality.

Inputs and Ensembles

- Provenance sidecar: results/uncertainty_pretest_provenance.json

- Score parquets: full, full_rescore, and 9 layer-ablation variants.

- Selection rates: 1/64, 1/32, 1/16, 1/8, 1/4

ensemble	role	members
mild	strict	full, mid2, top1
mid	direction_only	full, mid2, mid4, top1
wide	sanity_bound	full, top1, top2, mid2, bot2, top4, mid4, bot4, top6, skip
mild_no_full	diagnostic_only	mid2, top1

Noise Floor

full vs full_rescore: max absolute color difference = 0, mean = 0, fraction > 1e-6 = 0. This confirms that the spread analyzed below is structural ablation spread, not a clean posterior sample or stochastic scoring noise.

Registered Decision Rule

Positive iff, on mild, Test (a) threshold-band mean sigma exceeds deep-tail mean sigma with the difference CI excluding zero; Test (b) score-matched enriched sigma is lower than pure-C4 sigma with the difference CI excluding zero; and Test (c) the best positive-lambda retention at 1/64 beats lambda=0 with CI excluding zero and no same-rate pure-C4 recall loss. mid must have the same direction, but its CIs do not need to exclude zero.

Decision Summary

Table

condition	result
Test (a) mild strict	pass
Test (b) mild strict	pass
Test (c) mild strict	fail
mid same direction	fail
go/no-go call	weak_or_ambiguous

Test (a): Does Spread Concentrate Near the Boundary?

Table

ensemble	threshold - deep sigma	95% CI	threshold / deep sigma	95% CI
mild	0.0090	[0.0065, 0.0115]	1.1897	[1.1317, 1.2534]
mid	-0.0054	[-0.0084, -0.0025]	0.9377	[0.9061, 0.9701]
wide	0.0420	[0.0386, 0.0449]	1.1650	[1.1501, 1.1784]
mild_no_full	0.0119	[0.0091, 0.0151]	1.2350	[1.1721, 1.3117]

Spread vs full-score percentile: mild

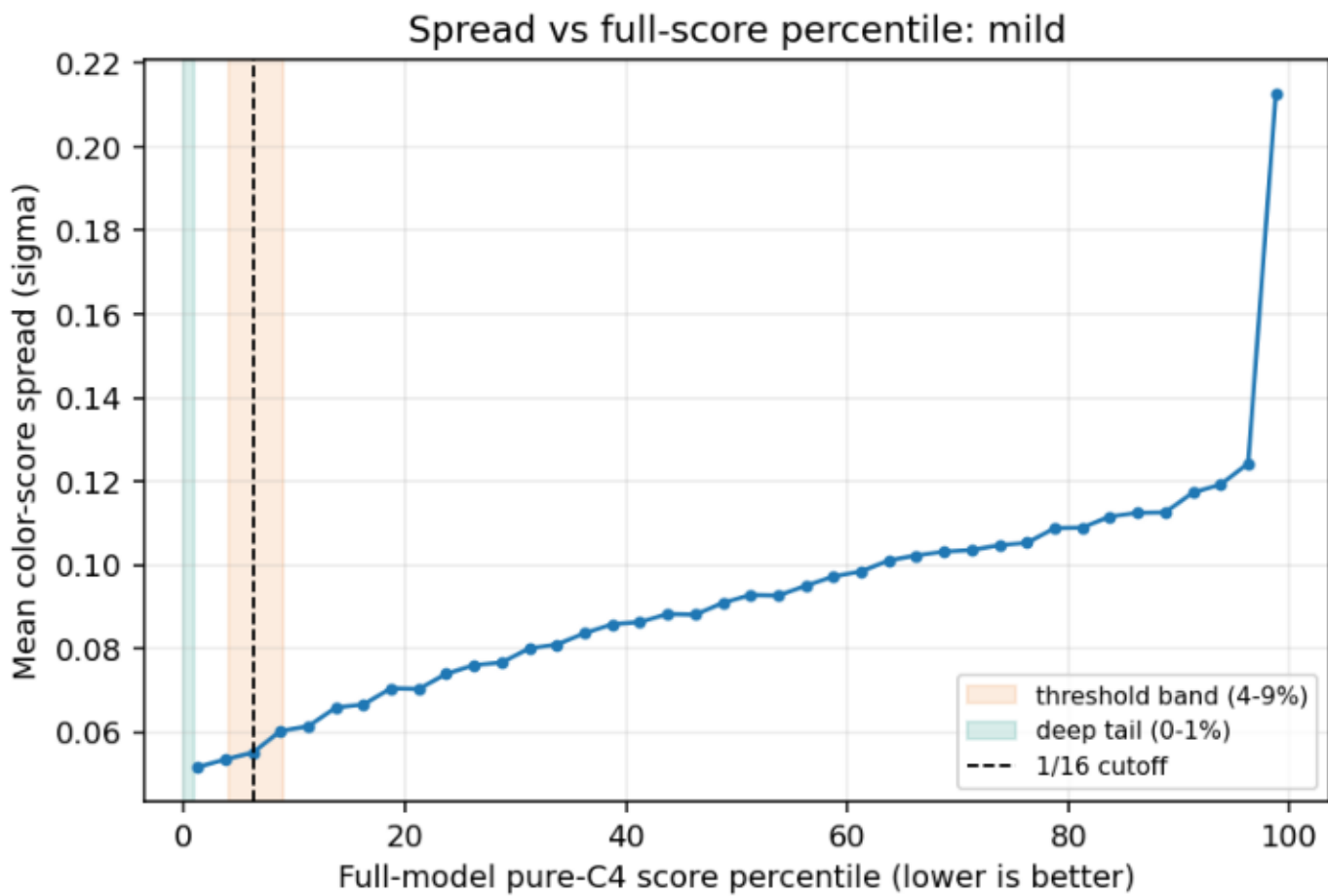


Figure: spread vs full-score percentile for mild. The shaded green band is the bottom 1% deep tail, the orange band is the 4-9% threshold region around the 1/16 cutoff, and the dashed line marks 6.25%.

Spread vs full-score percentile: mid

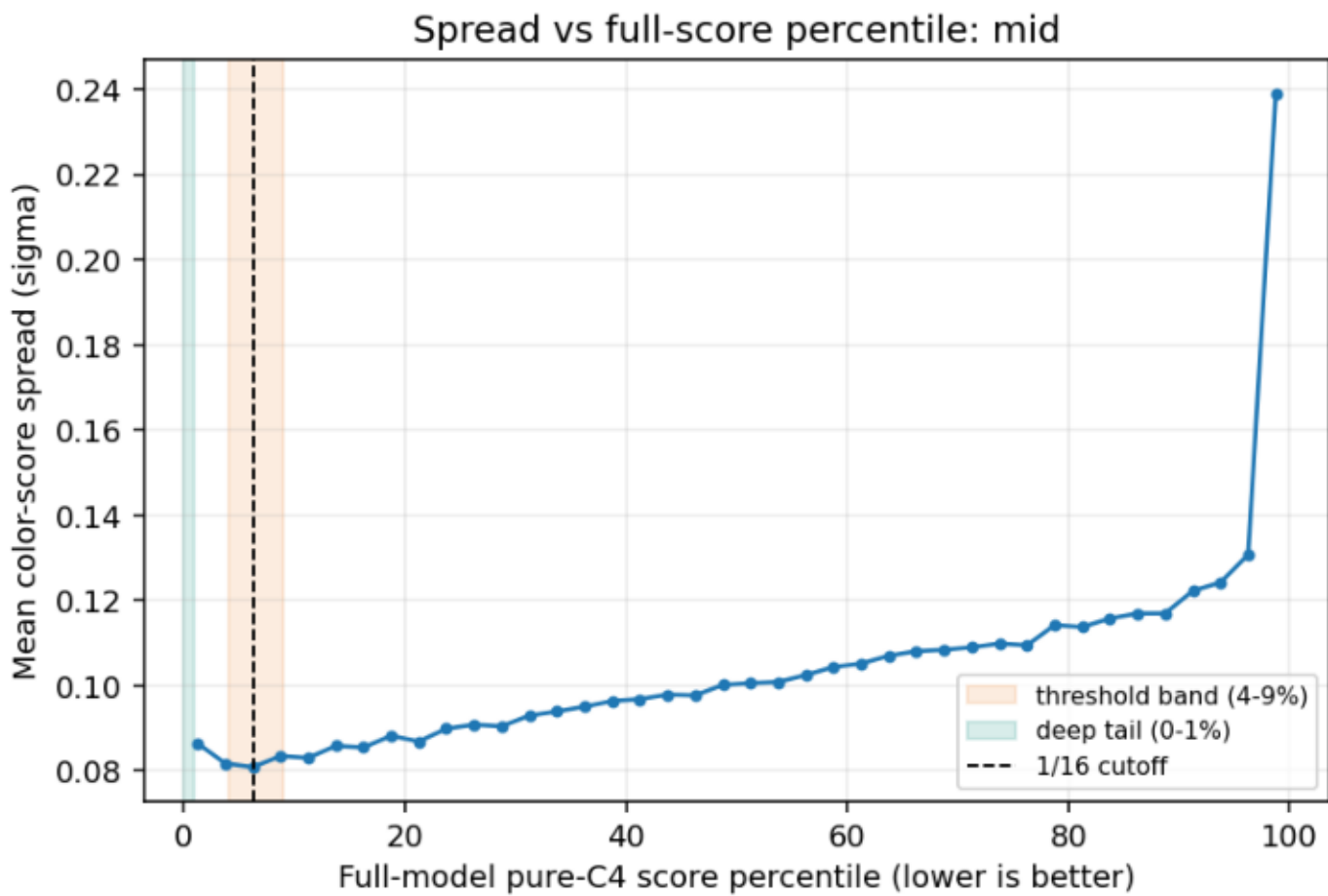


Figure: spread vs full-score percentile for mid. The shaded green band is the bottom 1% deep tail, the orange band is the 4-9% threshold region around the 1/16 cutoff, and the dashed line marks 6.25%.

Spread vs full-score percentile: wide

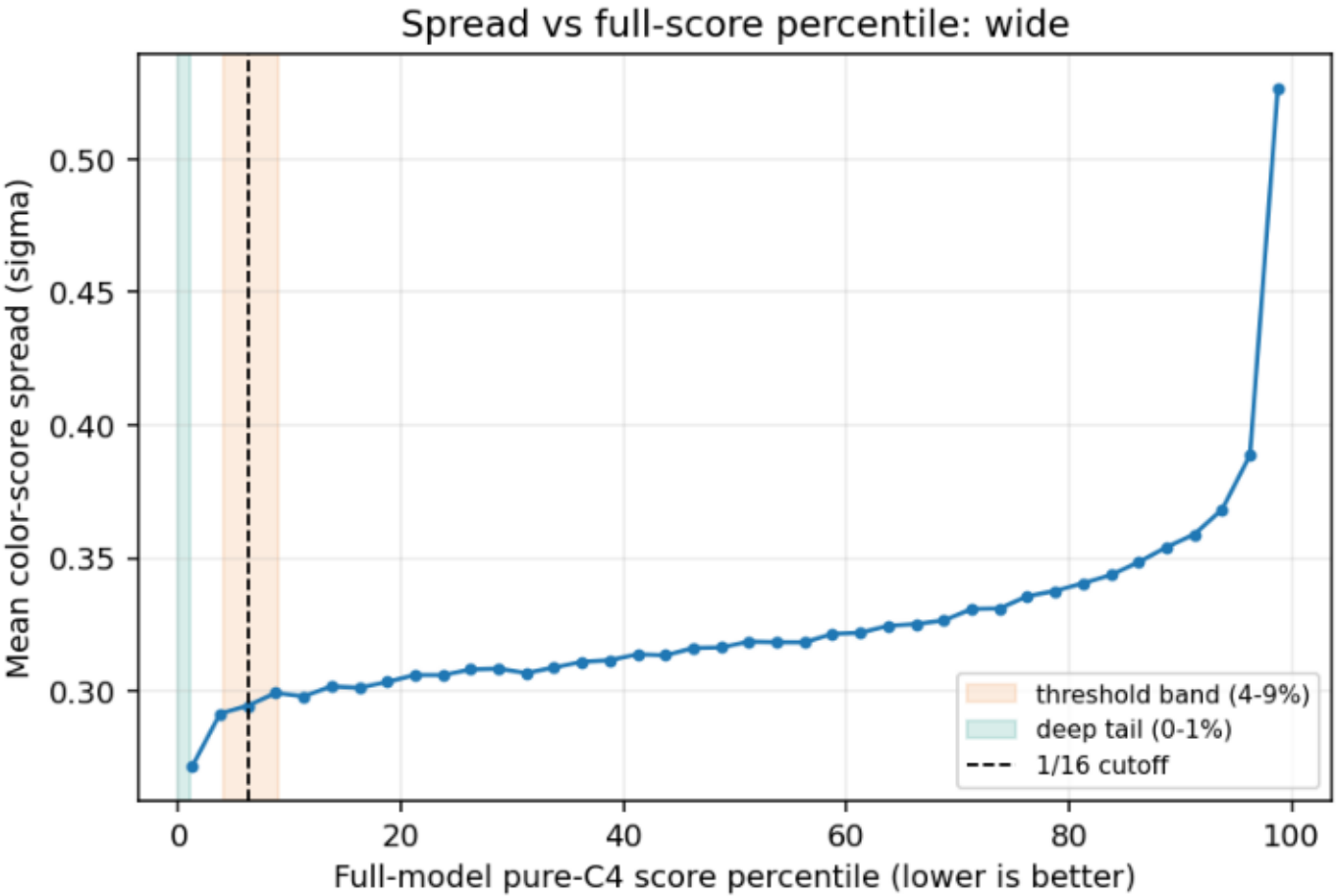


Figure: spread vs full-score percentile for wide. The shaded green band is the bottom 1% deep tail, the orange band is the 4-9% threshold region around the 1/16 cutoff, and the dashed line marks 6.25%.

Test (b): Is the Enriched Good Tail Low-Spread After Score Spread vs full-score percentile: mild_no_full Matching?

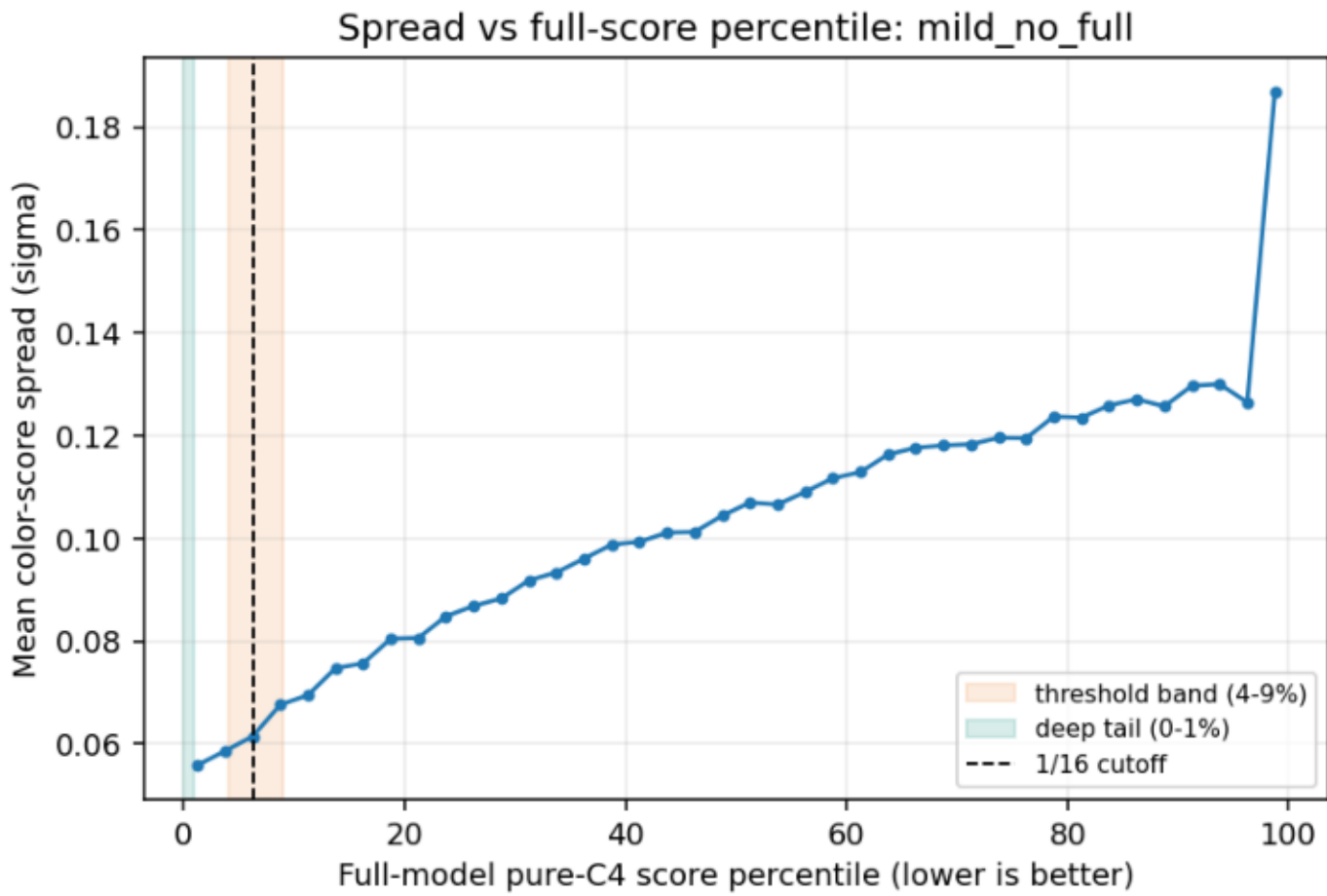


Figure: spread vs full-score percentile for mild_no_full. The shaded green band is the bottom 1% deep tail, the orange band is the 4-9% threshold region around the 1/16 cutoff, and the dashed line marks 6.25%.

Table

ensemble	band	enriched - pure matched sign	95% CI	matched n
mild	below_mu_1_16_pure_cutoff	-0.0044	[-0.0060, -0.0029]	2439
mild	below_mu_1pct_pure_cutoff	-0.0048	[-0.0069, -0.0028]	1000
mid	below_mu_1_16_pure_cutoff	0.0017	[-0.0001, 0.0036]	2538
mid	below_mu_1pct_pure_cutoff	-0.0057	[-0.0081, -0.0032]	1000
wide	below_mu_1_16_pure_cutoff	-0.0219	[-0.0240, -0.0198]	2755
wide	below_mu_1pct_pure_cutoff	-0.0167	[-0.0200, -0.0135]	1000
mild_no_full	below_mu_1_16_pure_cutoff	-0.0105	[-0.0123, -0.0086]	2572
mild_no_full	below_mu_1pct_pure_cutoff	-0.0075	[-0.0103, -0.0049]	1000

Score-matched sigma: mild

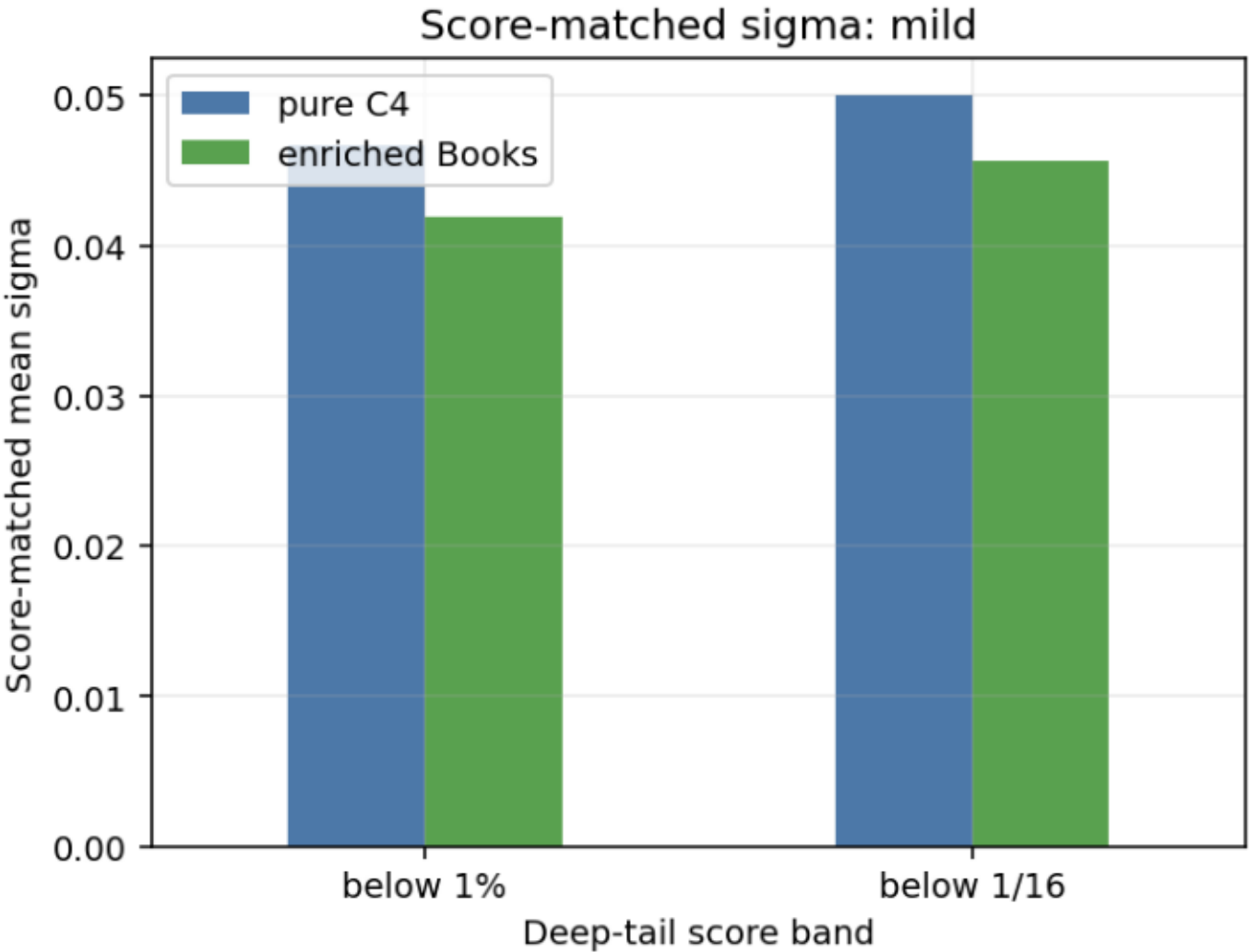


Figure: score-matched sigma for mild. Bars compare matched pure-C4 and enriched Books mean sigma inside deep-tail score bands. Negative enriched-minus-pure differences support the claim that known-good rows are stably low-scoring beyond score level alone.

Score-matched sigma: mid

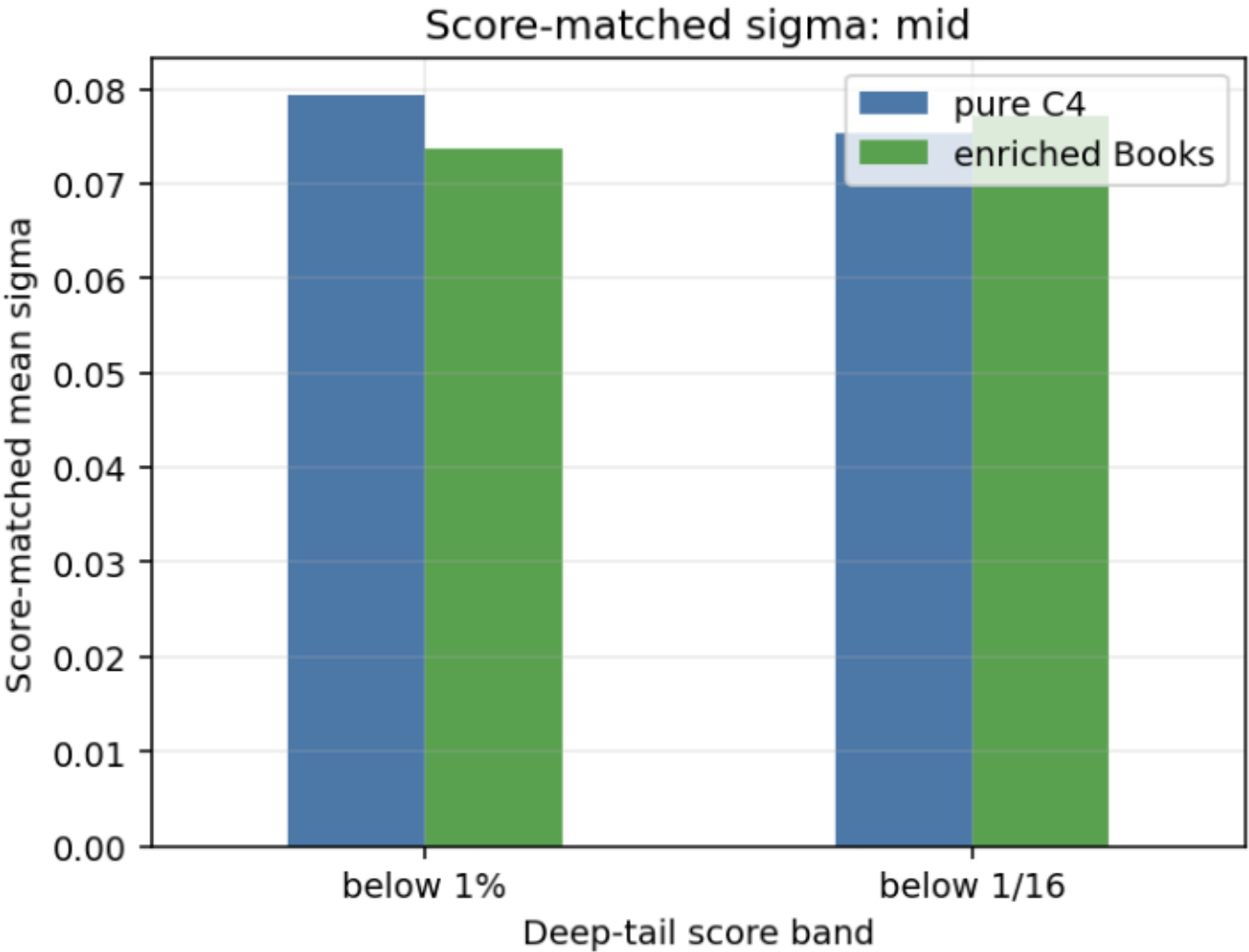


Figure: score-matched sigma for mid. Bars compare matched pure-C4 and enriched Books mean sigma inside deep-tail score bands. Negative enriched-minus-pure differences support the claim that known-good rows are stably low-scoring beyond score level alone.

Score-matched sigma: wide

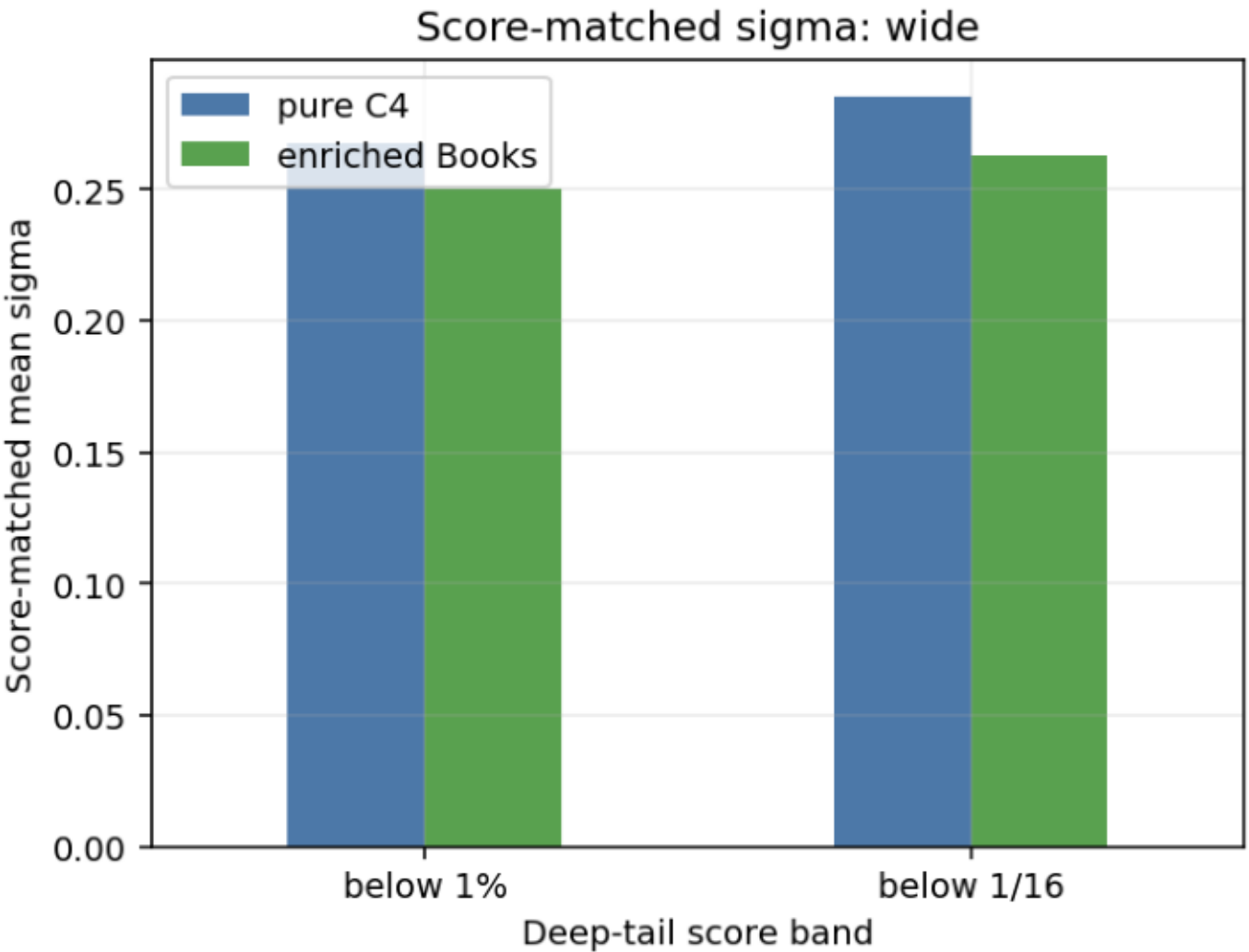


Figure: score-matched sigma for wide. Bars compare matched pure-C4 and enriched Books mean sigma inside deep-tail score bands. Negative enriched-minus-pure differences support the claim that known-good rows are stably low-scoring beyond score level alone.

Test (c): Does the Variance Term Carry Quality Signal?

Score-matched sigma: mild_no_full

ensemble	best positive lambda at 1/6	retention diff vs lambda=0	95% CI	pure-C4 recall diff vs lambda=0
mild	0.25	-0.0022	[-0.0056, 0.0012]	-0.0006
mid	0.25	0.0008	[-0.0034, 0.0048]	-0.0154
wide	1	0.0236	[0.0140, 0.0326]	0.0301
mild_no_full	0.5	0.0120	[0.0062, 0.0178]	0.0058

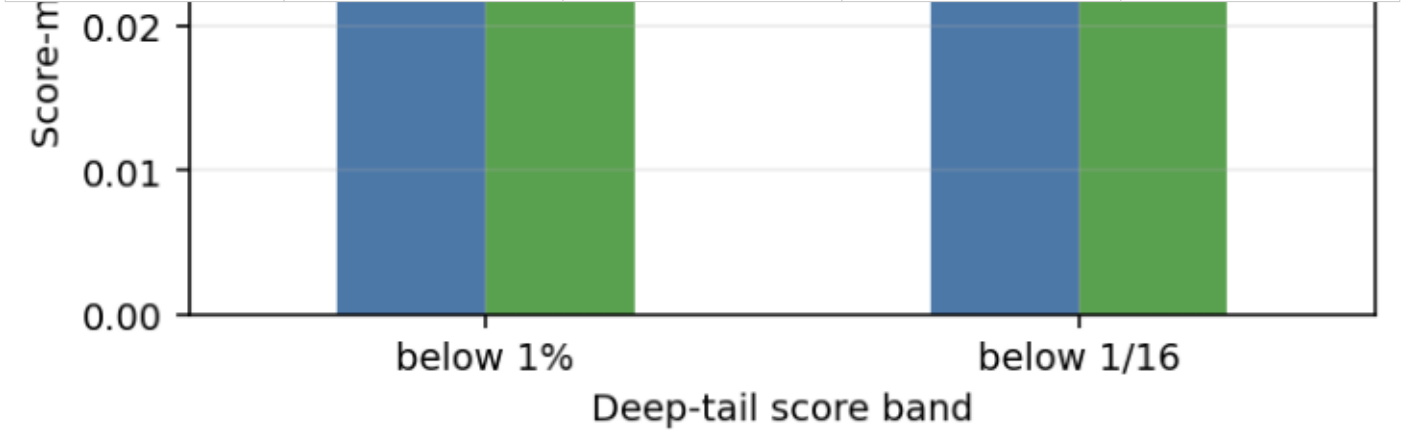


Figure: score-matched sigma for mild_no_full. Bars compare matched pure-C4 and enriched Books mean sigma inside deep-tail score bands. Negative enriched-minus-pure differences support the claim that known-good rows are stably low-scoring beyond score level alone.

Retention vs lambda: mild

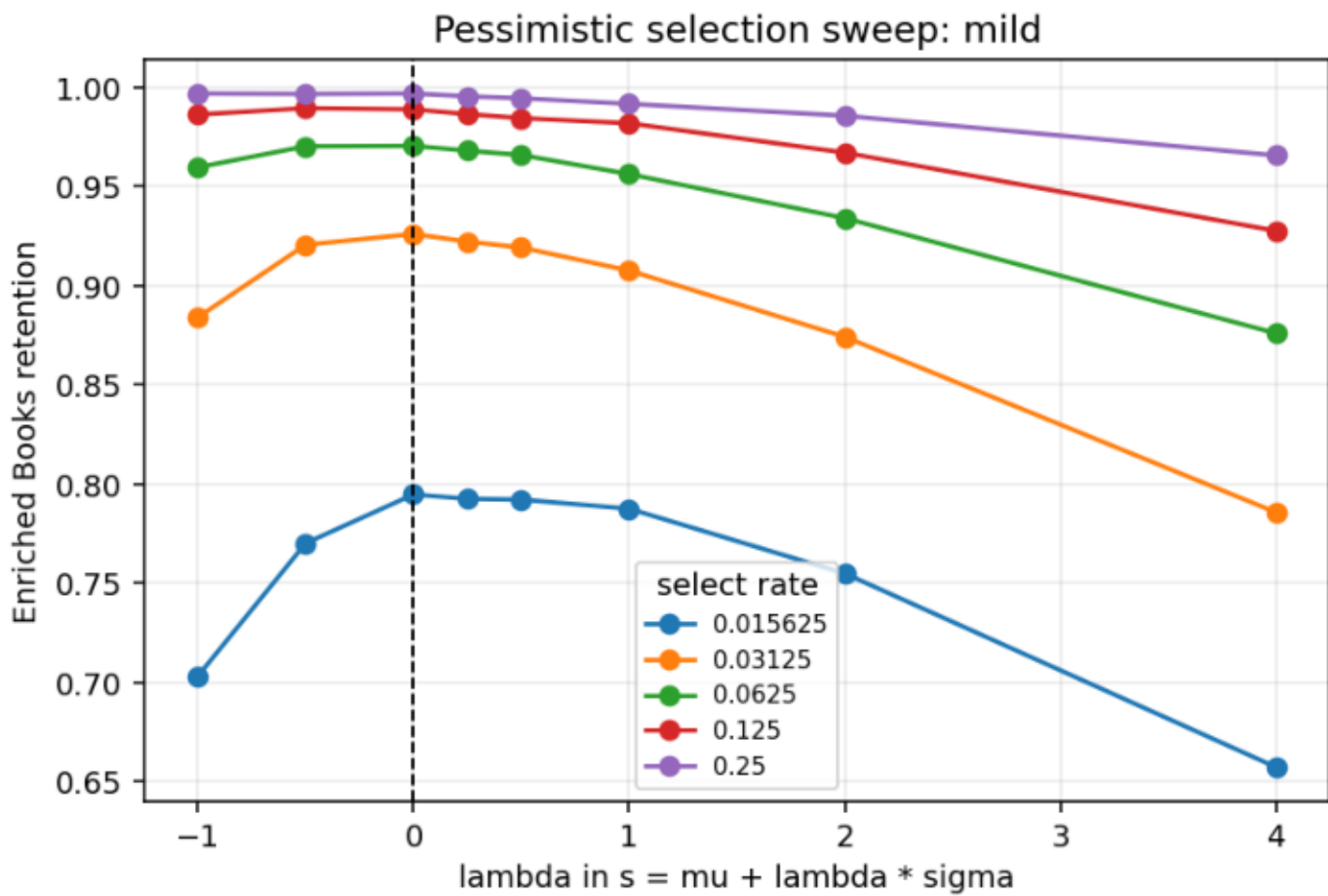


Figure: retention vs lambda for mild. Lambda=0 is the point-estimate ensemble ranking by mu. Positive lambda penalizes high sigma; negative lambda is the optimism control.

Retention vs recall: mild

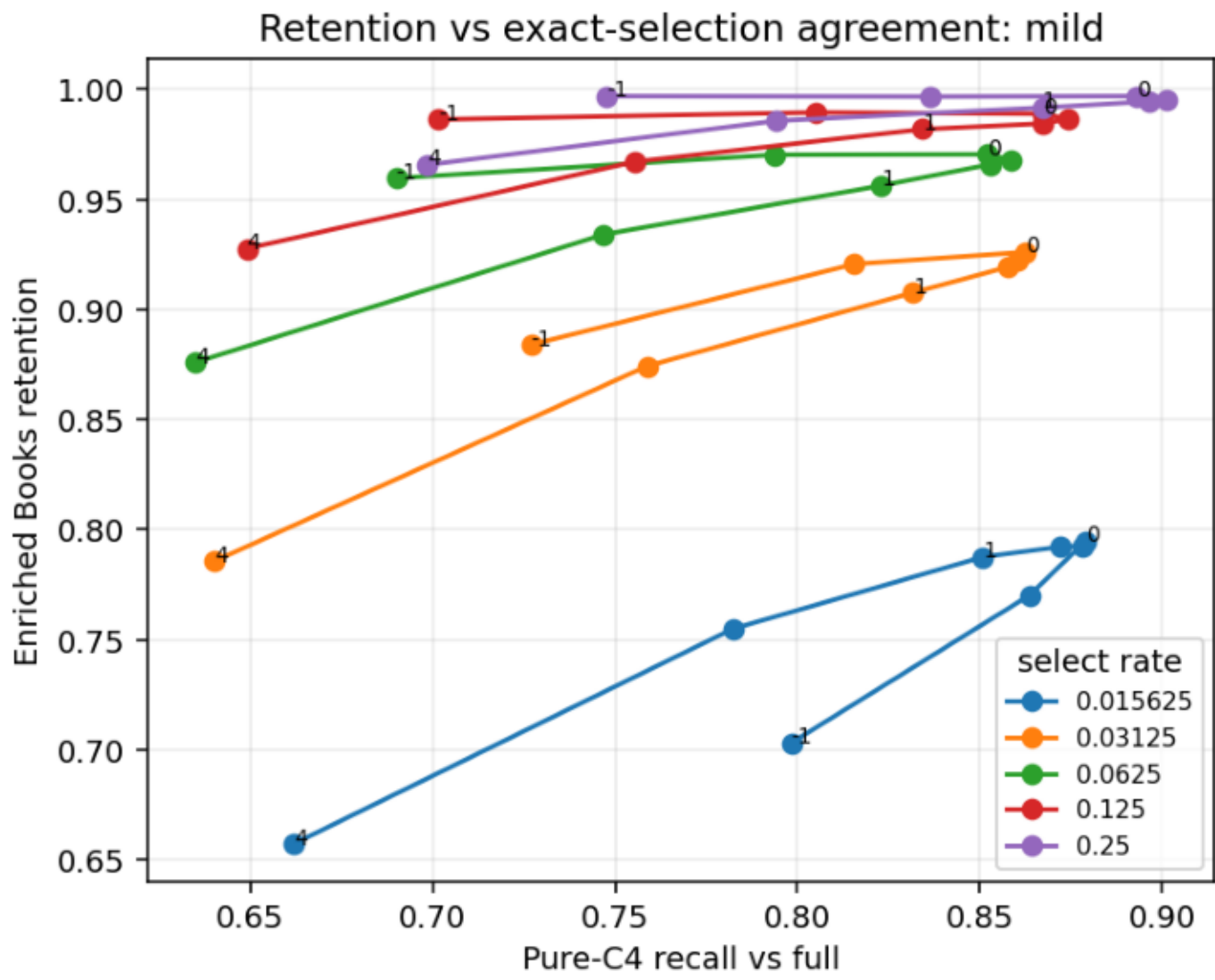


Figure: retention-vs-recall tradeoff for mild. This checks whether any enriched-retention gain is bought by losing agreement with the full pure-C4 selected set.

Retention vs lambda: mid

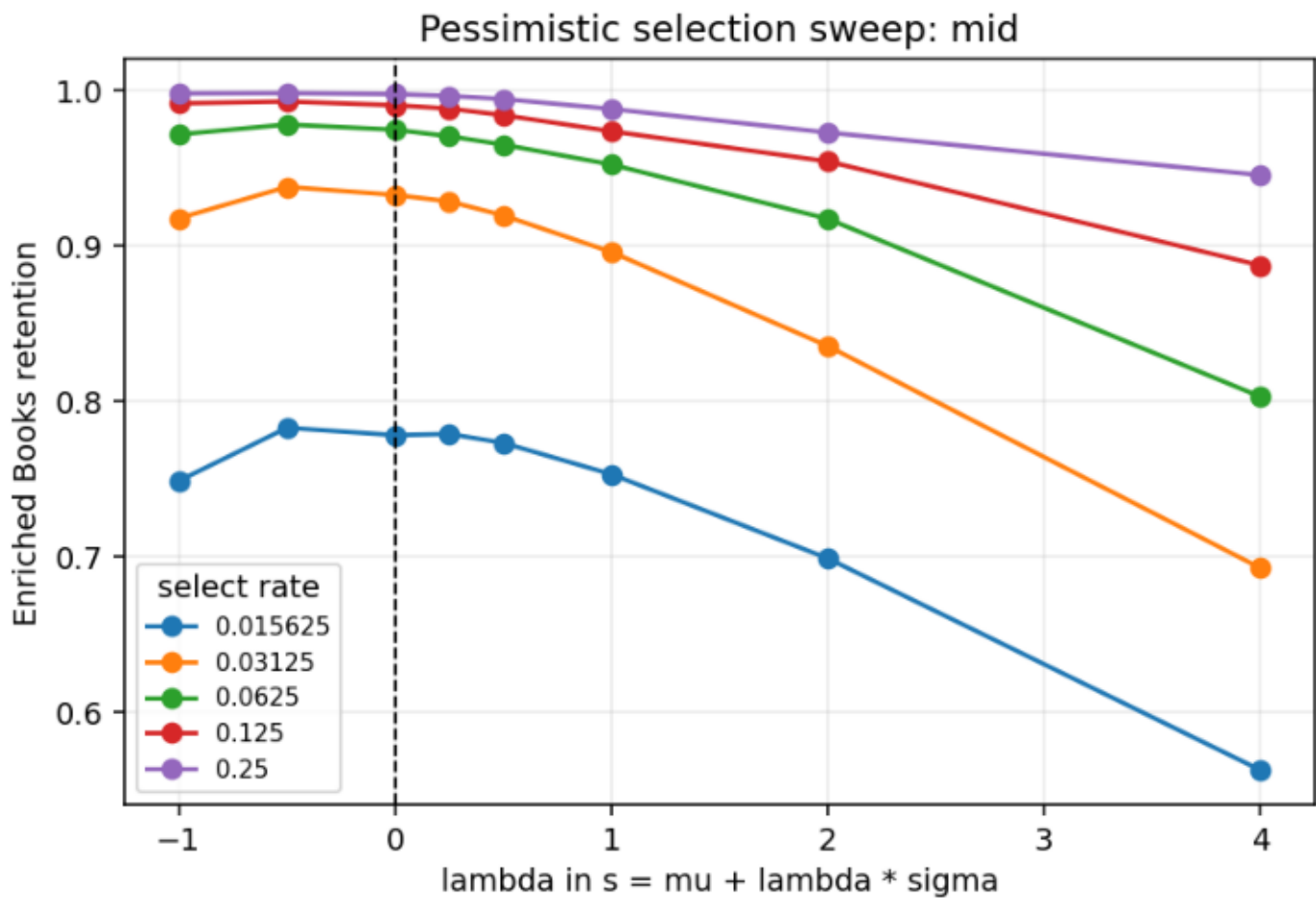


Figure: retention vs lambda for mid. Lambda=0 is the point-estimate ensemble ranking by mu. Positive lambda penalizes high sigma; negative lambda is the optimism control.

Retention vs recall: mid

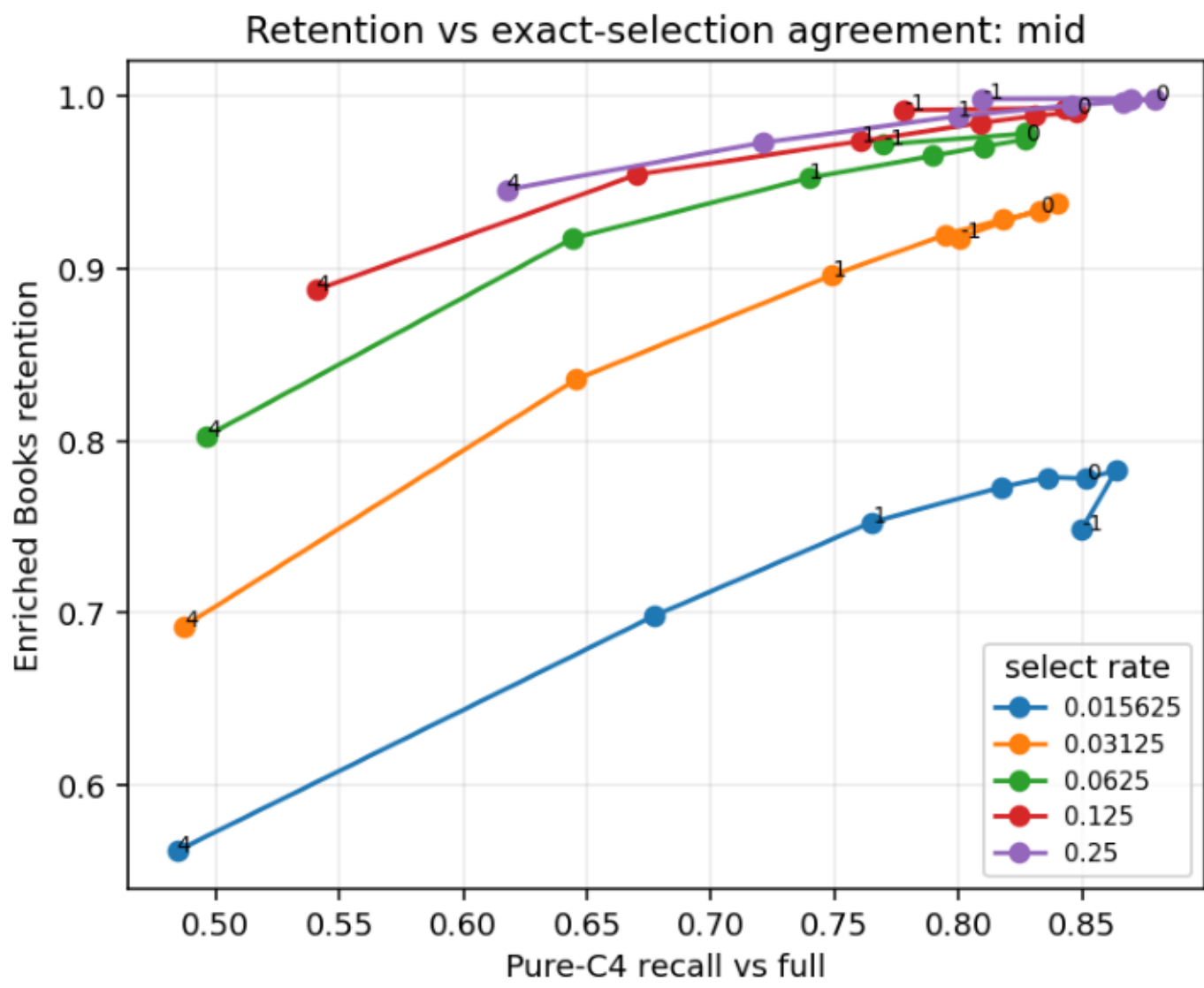


Figure: retention-vs-recall tradeoff for mid. This checks whether any enriched-retention gain is bought by losing agreement with the full pure-C4 selected set.

Retention vs lambda: wide

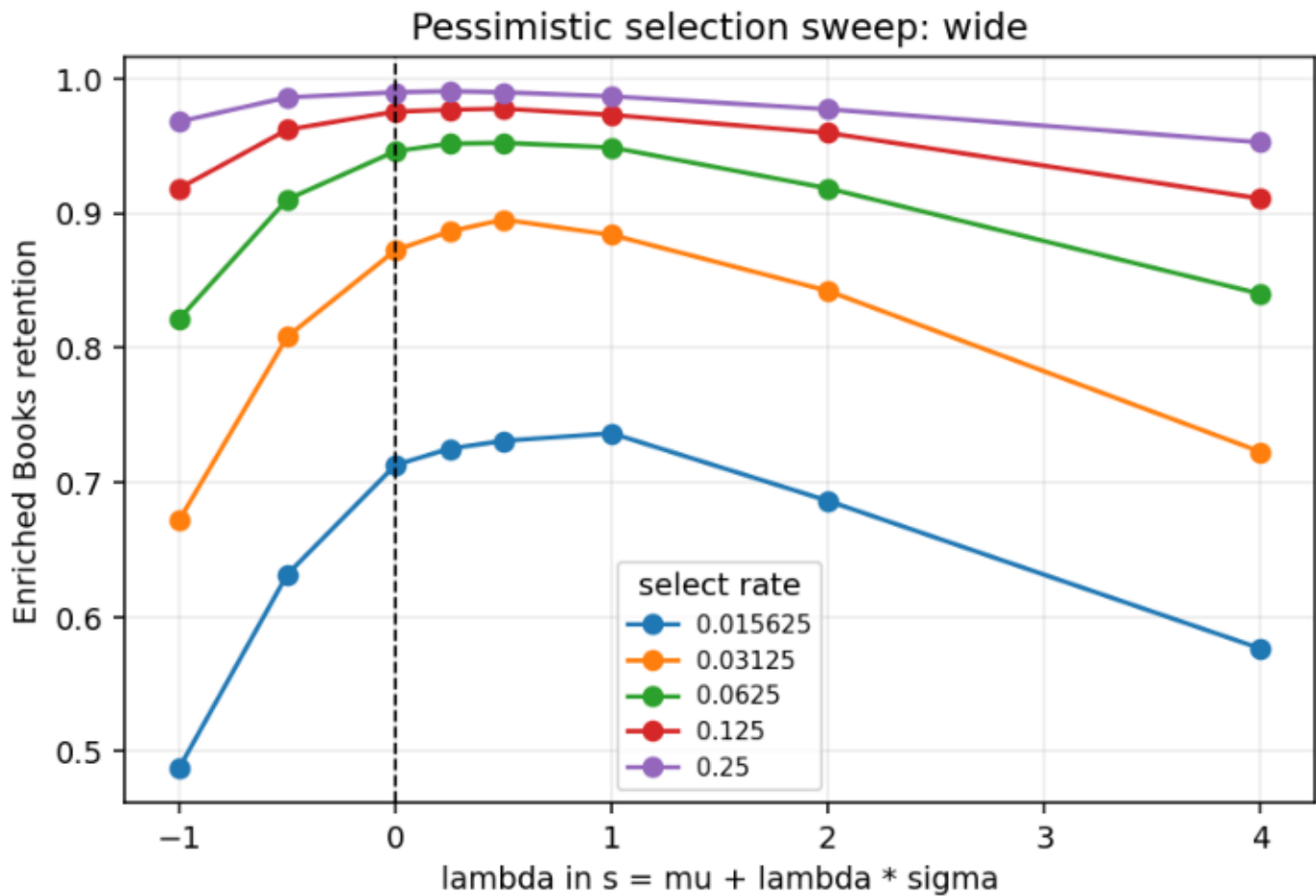


Figure: retention vs lambda for wide. Lambda=0 is the point-estimate ensemble ranking by mu. Positive lambda penalizes high sigma; negative lambda is the optimism control.

Retention vs recall: wide

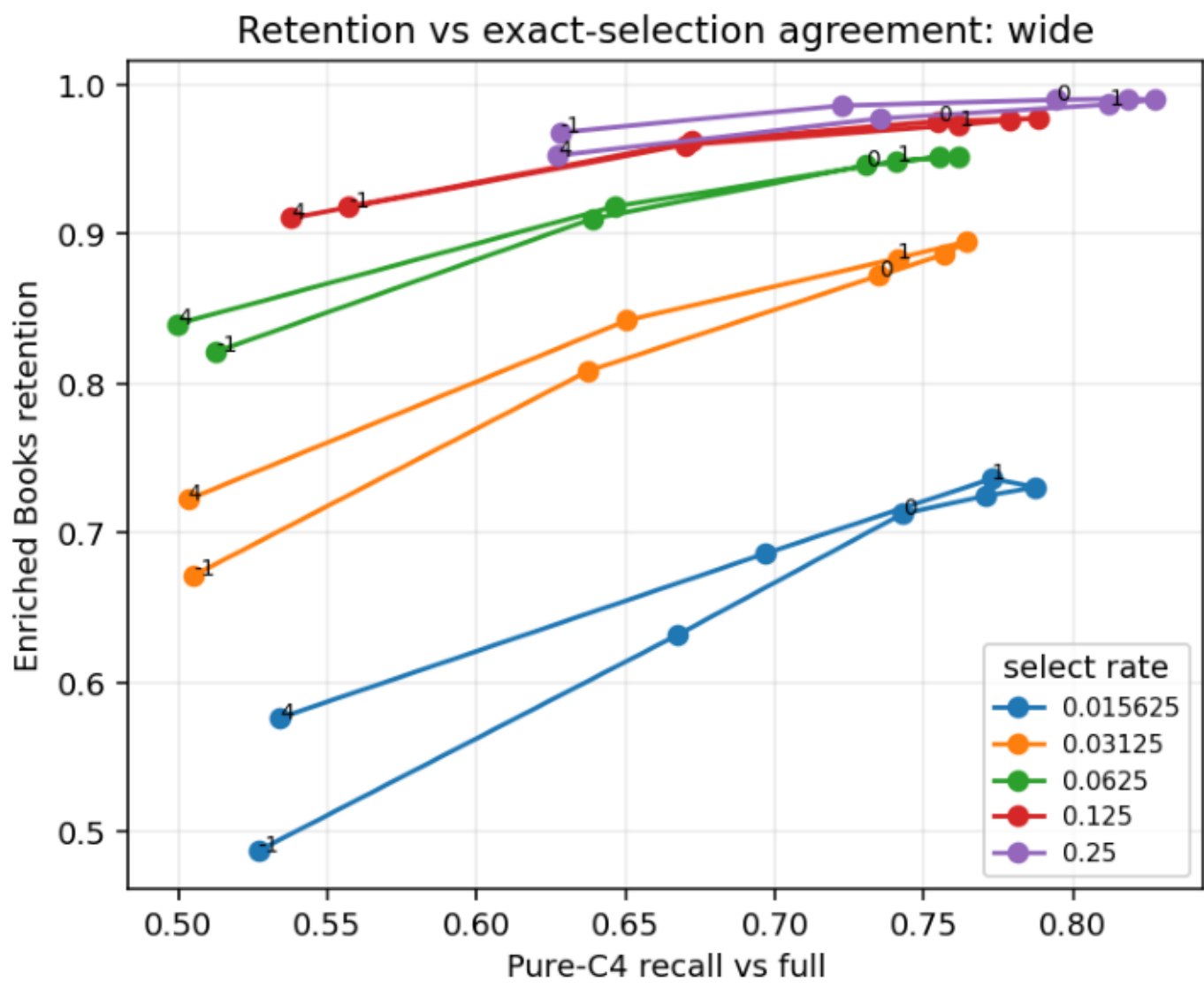


Figure: retention-vs-recall tradeoff for wide. This checks whether any enriched-retention gain is bought by losing agreement with the full pure-C4 selected set.

Retention vs lambda: mild_no_full

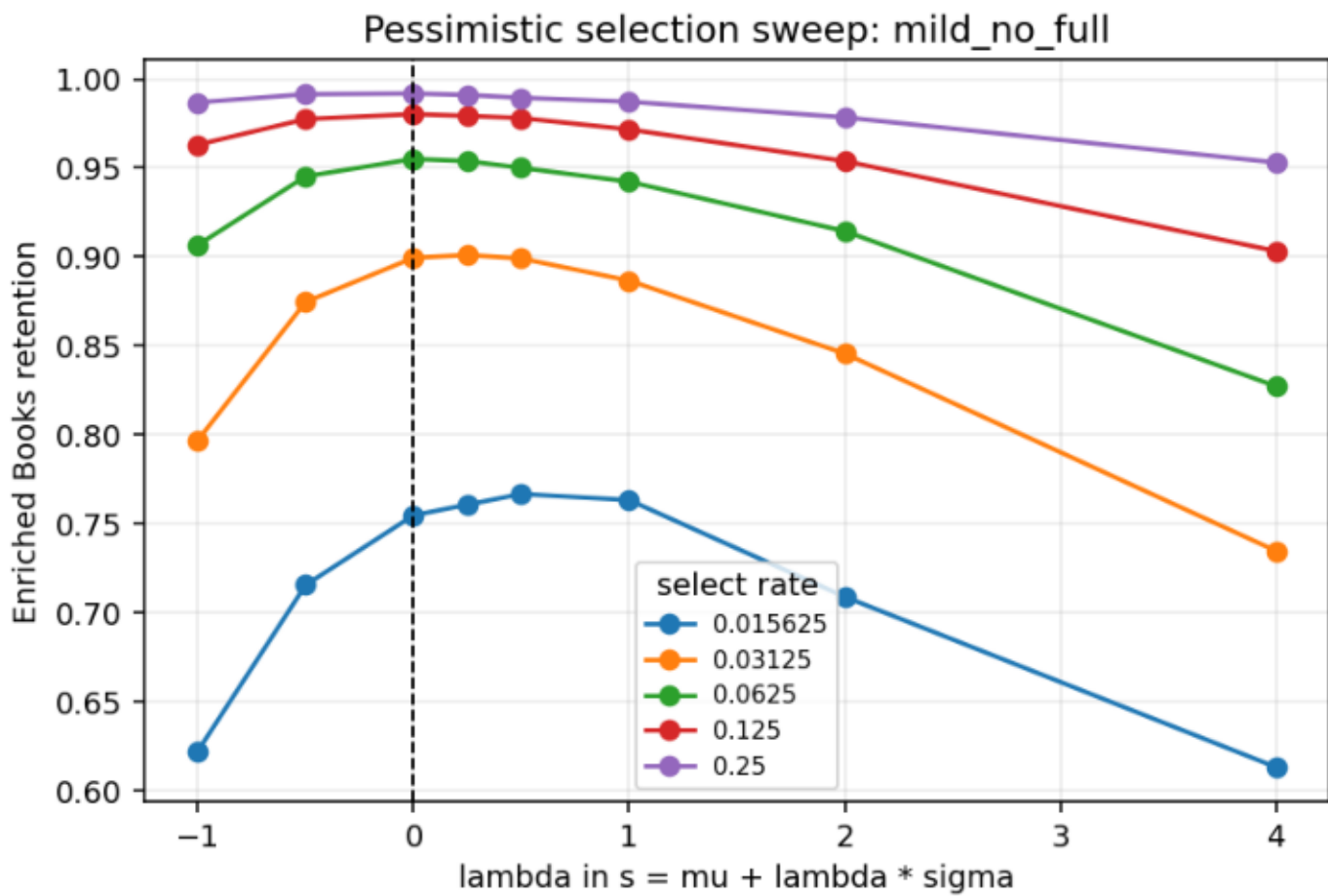


Figure: retention vs lambda for mild_no_full. Lambda=0 is the point-estimate ensemble ranking by mu. Positive lambda penalizes high sigma; negative lambda is the optimism control.

Interpretation

Retention Vs recall: mild_no_full

The proxy-disturbance pre-test finds heteroscedasticity signals but does not show that the sigma term improves selection under the strict rule. This argues against building the real ensemble immediately at 12-layer depth unless the goal is exploratory.

Reproducibility Appendix

Retention vs exact-selection agreement: mild_no_full

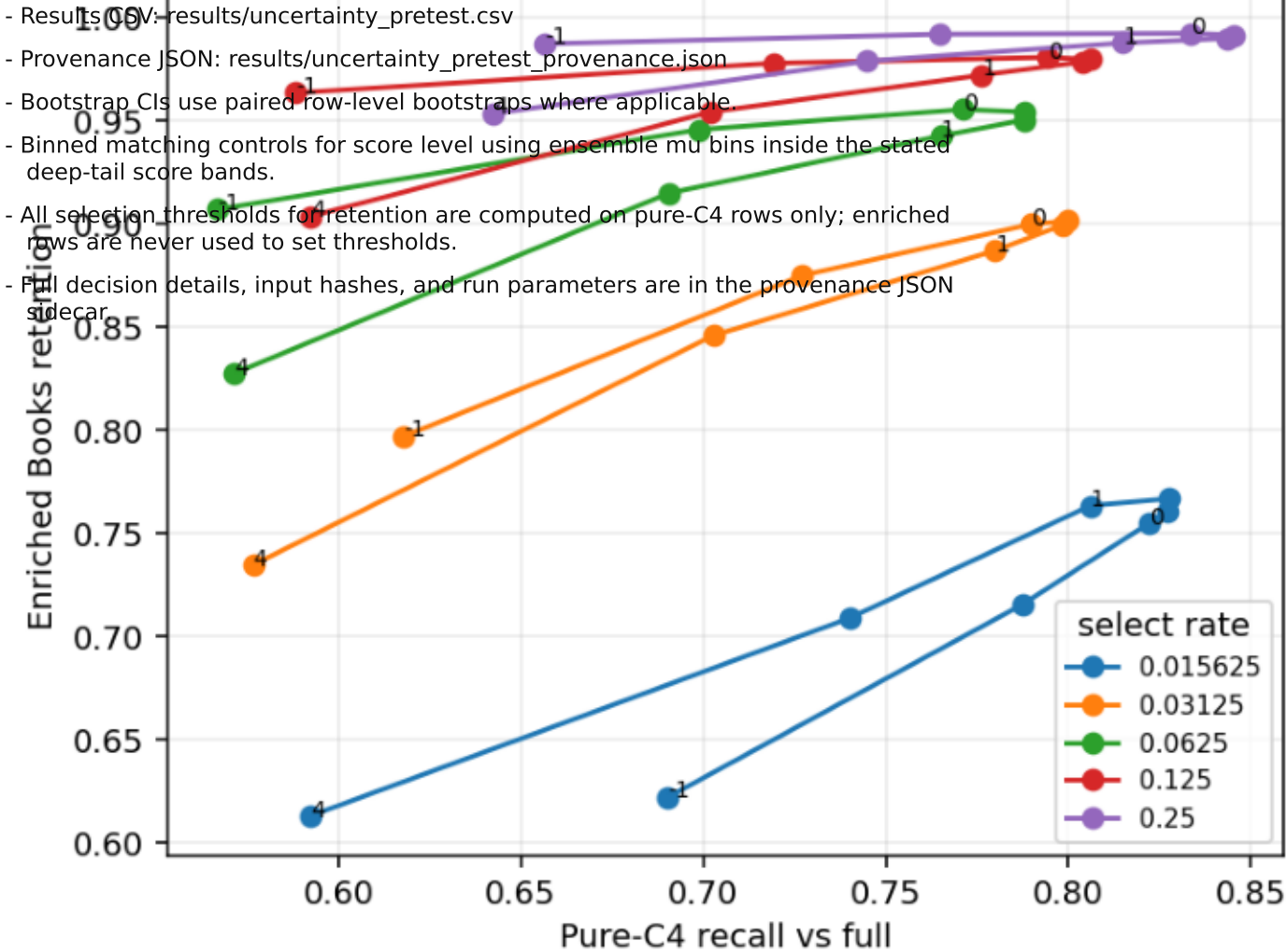


Figure: retention-vs-recall tradeoff for mild_no_full. This checks whether any enriched-retention gain is bought by losing agreement with the full pure-C4 selected set.